CSE Placement Cheat Sheet

Most Important Concepts for Computer Science Students Before Placements

1. Data Structures & Algorithms

• Arrays & Strings: Sliding Window, Two Pointer, Prefix Sum
• Linked Lists: Reverse, Detect Cycle, Merge Lists
• Stacks & Queues: Monotonic Stack, BFS Queue
• Trees: DFS, BFS, BST, Traversals, Lowest Common Ancestor
• Graphs: BFS, DFS, Dijkstra, Topological Sort, Union Find
• Dynamic Programming: Memoization, Tabulation, Knapsack, LIS
• Sorting: Merge Sort, Quick Sort, Heap Sort
• Complexities: Understand Time & Space Complexity

2. OOPs Concepts

• Encapsulation: Data hiding using classes
• Abstraction: Hide implementation details
• Inheritance: Reuse properties from parent class
• Polymorphism: Method Overloading & Overriding
• Constructor vs Method
• Interface vs Abstract Class
• Static vs Non-static
• Access Modifiers: public, private, protected

3. DBMS

• Normalization: 1NF, 2NF, 3NF, BCNF
• Primary Key vs Foreign Key
• Joins: INNER, LEFT, RIGHT, FULL
• Indexing improves query speed
• Transactions & ACID Properties
• SQL Basics: GROUP BY, HAVING, ORDER BY
• Difference between DELETE, DROP, TRUNCATE

4. Operating Systems

• Process vs Thread
• Deadlock Conditions
• CPU Scheduling Algorithms
• Paging vs Segmentation
• Virtual Memory
• Synchronization: Mutex, Semaphore
• Context Switching

5. Computer Networks

• OSI Model & TCP/IP Model
• HTTP vs HTTPS
• TCP vs UDP
• IP Address, DNS, MAC Address
• 3-Way Handshake
• Subnetting Basics
• Common Ports: 80, 443, 22

6. System Design Basics

• Load Balancer
• Caching
• Database Sharding
• CAP Theorem
• Scalability
• Monolith vs Microservices

7. Aptitude & Problem Solving

• Percentages, Ratios, Profit & Loss
• Probability & Permutations
• Speed, Distance, Time
• Logical Reasoning
• Puzzle Solving

8. HR & Interview Tips

• Prepare Projects Deeply
• Know Resume Thoroughly
• Explain Tech Stack Clearly
• Practice Behavioral Questions
• Communicate Clearly
• Think Aloud During Coding